ASR100B120B-T3A Datasheet

- 50 mm up to 305 mm travel per axis
- Up to 12 µm accuracy; 2 µm repeatability; 85 mm/s speed
- Easily adjustable non-contact hall sensor limit switches
- Supported by µManager and MetaMorph® Microscopy Automation & Image Analysis Software
- Designed for use with an X-MCC2 Series stepper motor controller or any 2-phase stepper motor controller
- With AutoDetect, the X-MCC controller configures its settings automatically for the connected peripheral
- Custom versions available

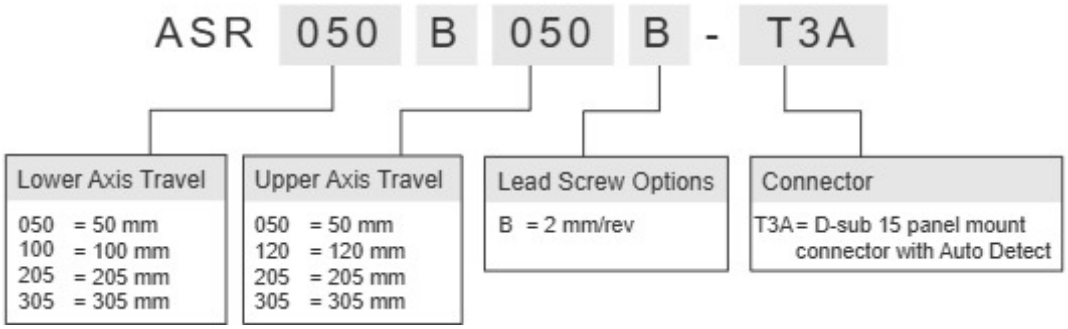
ASR Series Overview

Zaber's ASR series microscope stages combined with our X-MCC2 series multi-axis controllers are designed as replacements for manual stages on upright and inverted microscopes or for stand-alone operation as scanning stages. Set up is easy with AutoDetect. Once connected, the X-MCC controller will automatically detect and configure the ASR.

Mounting adaptors are available for breadboards and most common microscopes. Custom adaptors and plates are available upon request.

For more information visit: https://www.zaber.com/products/scanning-microscope-stages/ASR

ASR Series Part Numbering & Options



ASR100B120B-T3A Drawings

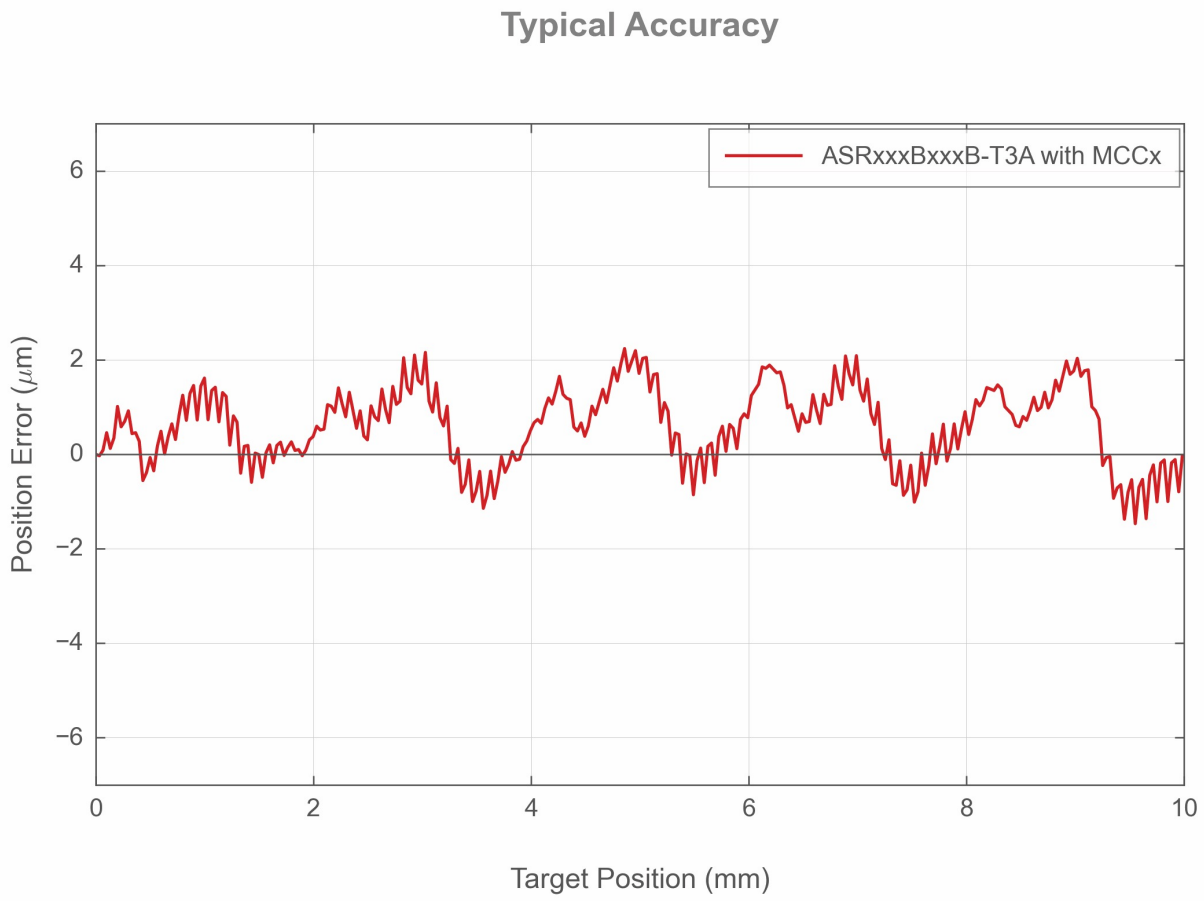
- ASR100B120B.png (Drawing for the ASR100B120B)

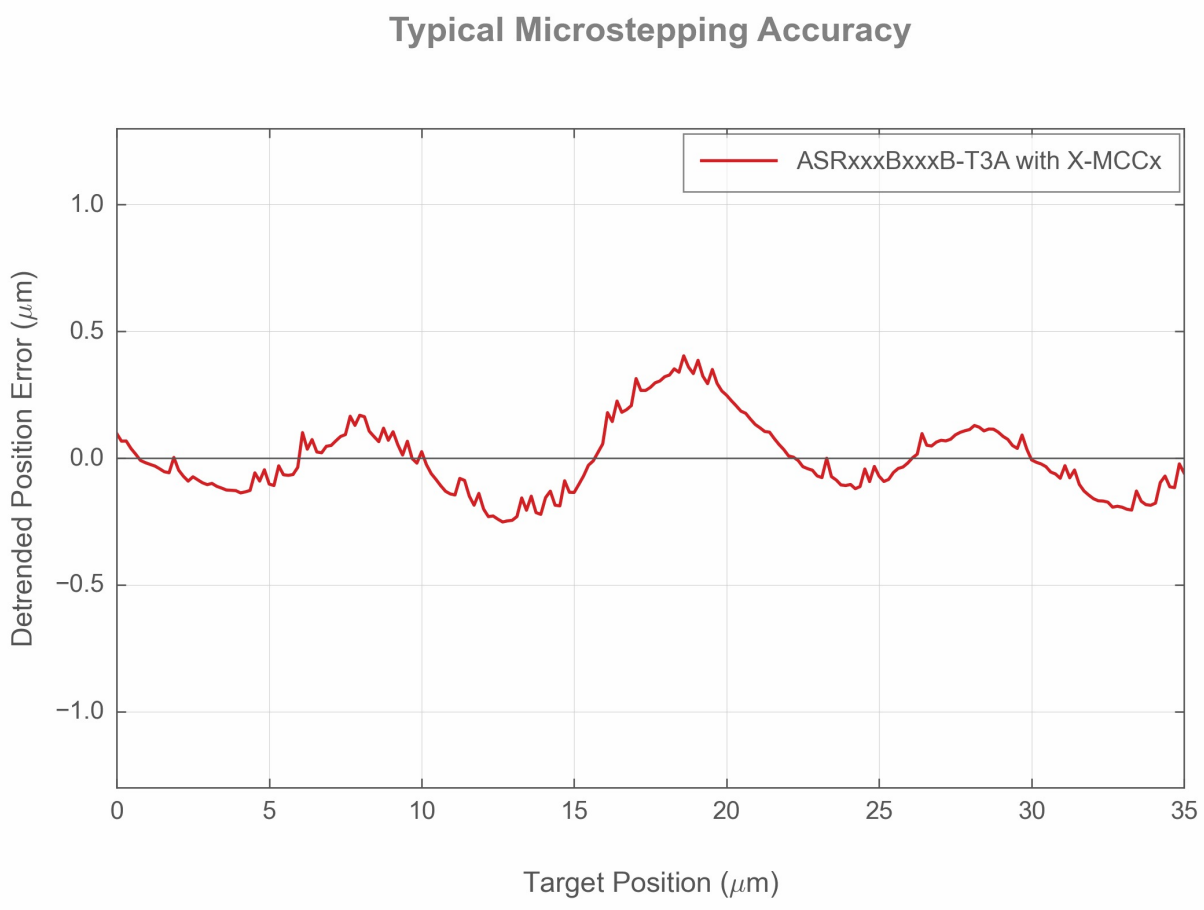
ASR100B120B-T3A Specifications

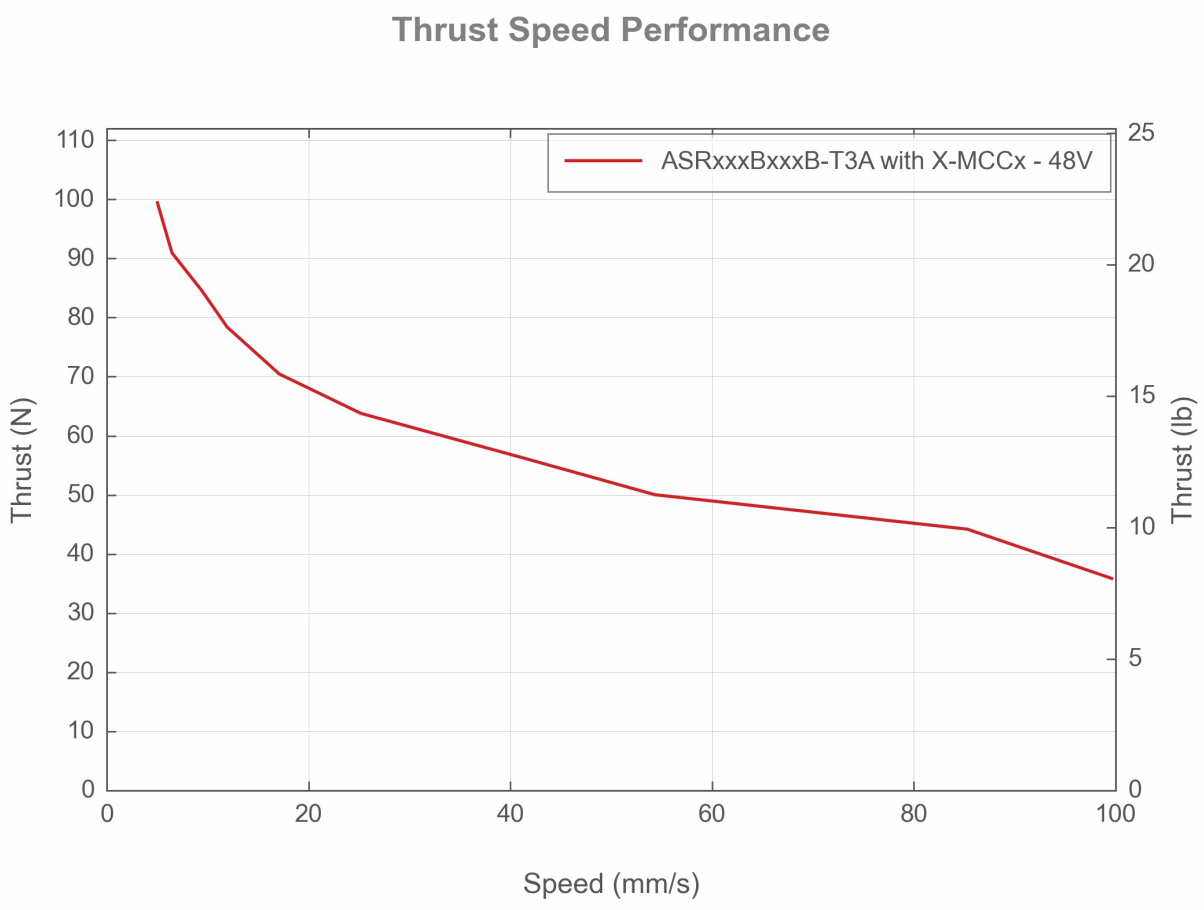
Microstep Size (Default Resolution)	**0.15625 µm**
Built-in Controller	No
Recommended Controller	X-MCC2 (48 V) Recommended
AutoDetect	Yes
Lower Travel Range	100 mm (3.937")
Upper Travel Range	120 mm (4.724")
Accuracy (unidirectional)	40 µm (0.001575")
Repeatability	< 2 µm (< 0.000079")
Flatness	20 µm (0.0008")
Backlash	< 4 µm (< 0.000157")
Maximum Speed	85 mm/s (3.346"/s)
Minimum Speed	0.000095 mm/s (0.000004"/s)
Speed Resolution	0.000095 mm/s (0.000004"/s)
Peak Thrust	95 N (21.3 lb)
Maximum Continuous Thrust	95 N (21.3 lb)
Maximum Centered Load	100 N (22.4 lb)
Guide Type	Crossed roller bearing
Pitch	0.02° (0.349 mrad)
Roll	0.02° (0.349 mrad)
Yaw	0.005° (0.087 mrad)
Linear Motion Per Motor Rev	2.00 mm (0.079")
Motor Steps Per Rev	200
Motor Type	Stepper (2 phase)
Motor Rated Current	1500 mA/phase
Motor Winding Resistance	2.05 ohms/phase
Inductance	1 mH/phase
Motor Connection	D-sub 15
Default Resolution	1/64 of a step
Mechanical Drive System	Precision lead screw
Limit or Home Sensing	Magnetic, adjustable home and away sensors
Axes of Motion	2

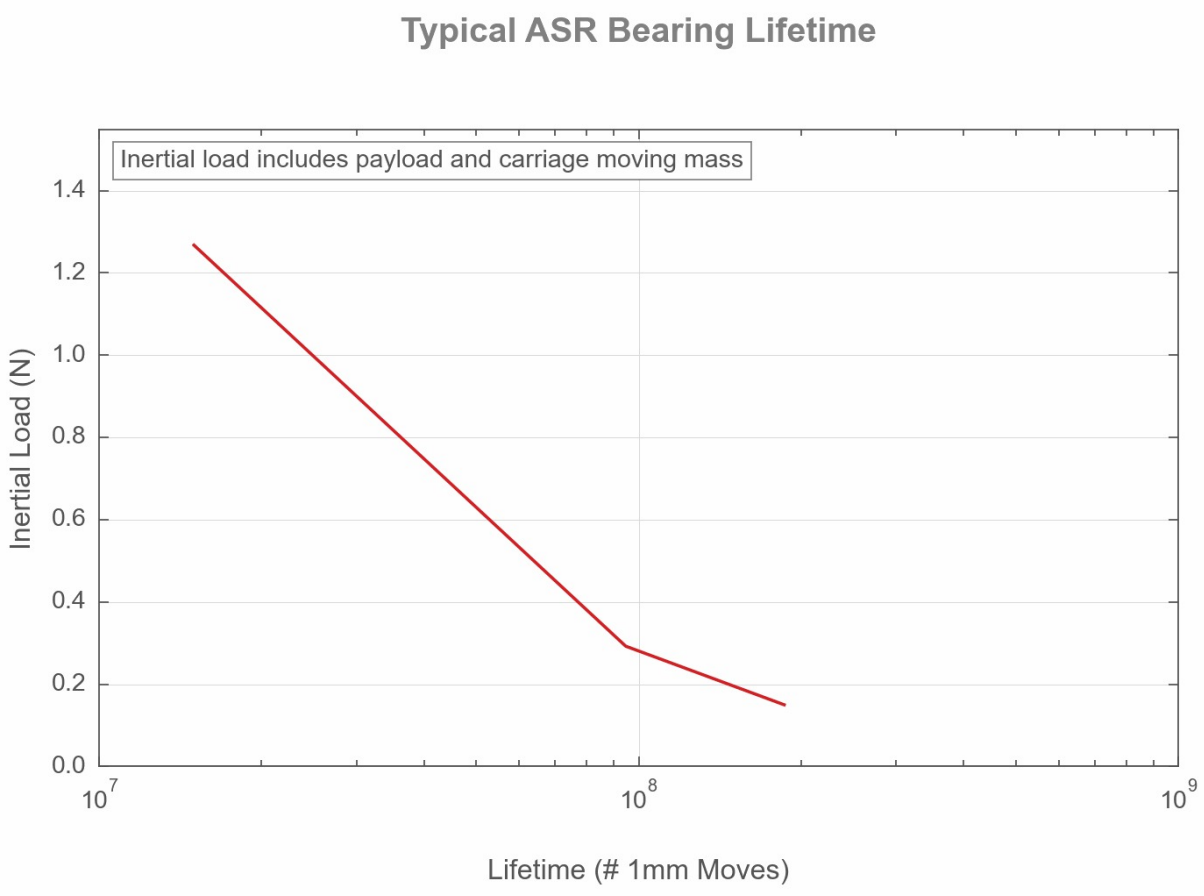
Microstep Size (Default Resolution)	**0.15625 µm**
Mounting Interface	Separate mounting adaptors available
Operating Temperature Range	0 to 50 °C
RoHS Compliant	Yes
CE Compliant	Yes
Vacuum Compatible	No
Weight	3 kg (6.614 lb)

ASR Series Charts




Typical Microstepping Accuracy
ASRxxxBxxxB-T3A with X-MCCx
Detrended Position Error (μm)
1.0
0.5
0.0
−0.5
−1.0
0
5
10
15
20
25
30
35
Target Position (μm)


Thrust Speed Performance
ASRxxxBxxxB-T3A with X-MCCx - 48V
Thrust (N)
Thrust (lb)
Speed (mm/s)
0
10
20
30
40
50
60
70
80
90
100
110
0
5
10
15
20
25
0
20
40
60
80
100



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